

Filing Lens

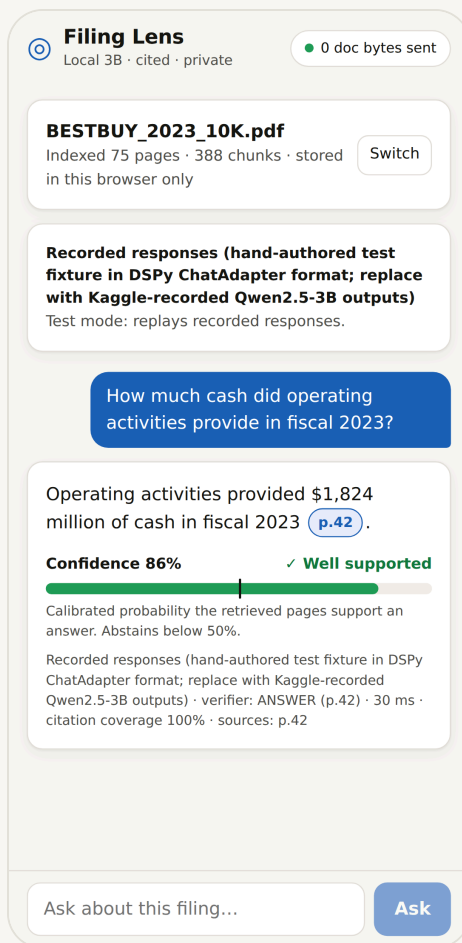
Cited answers from company filings, from a local 3B model (Qwen 2.5, runs on the phone)

Retail investors are handed 100-page 10-Ks. Chatbots answer anyway, with no page reference, and the filing gets uploaded to someone else's server.

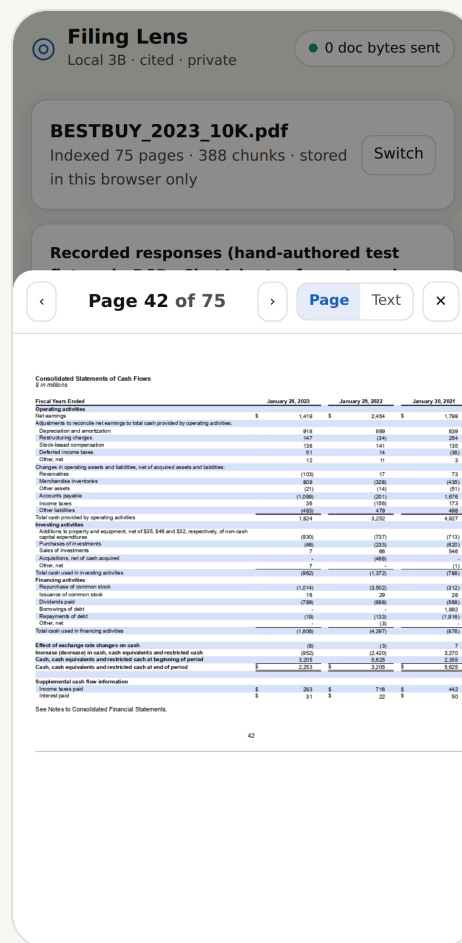
FinanceBench (Islam et al., 2023, arXiv:2311.11944): GPT-4-Turbo with a retrieval system **incorrectly answered or refused 81%** of 150 open-book filing questions.

Problem: wrong answers look the same as right ones, and privacy is the price of asking.

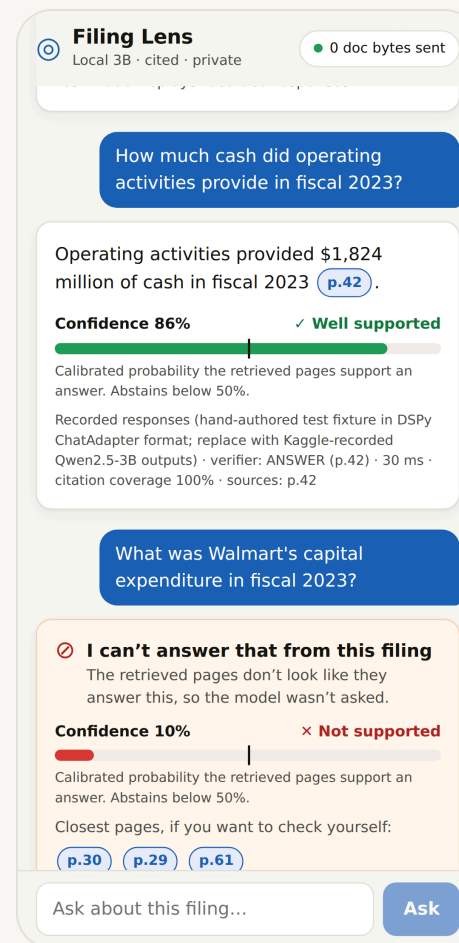
Demo



Cited answer + calibrated confidence



Citation chip opens the page



Wrong-company question → abstain

Recorded in headless Chromium at Pixel-7 size with recorded model responses (the build machine can't download weights). Phone video: [\[ASK: phone recording link\]](#) .

Privacy proof

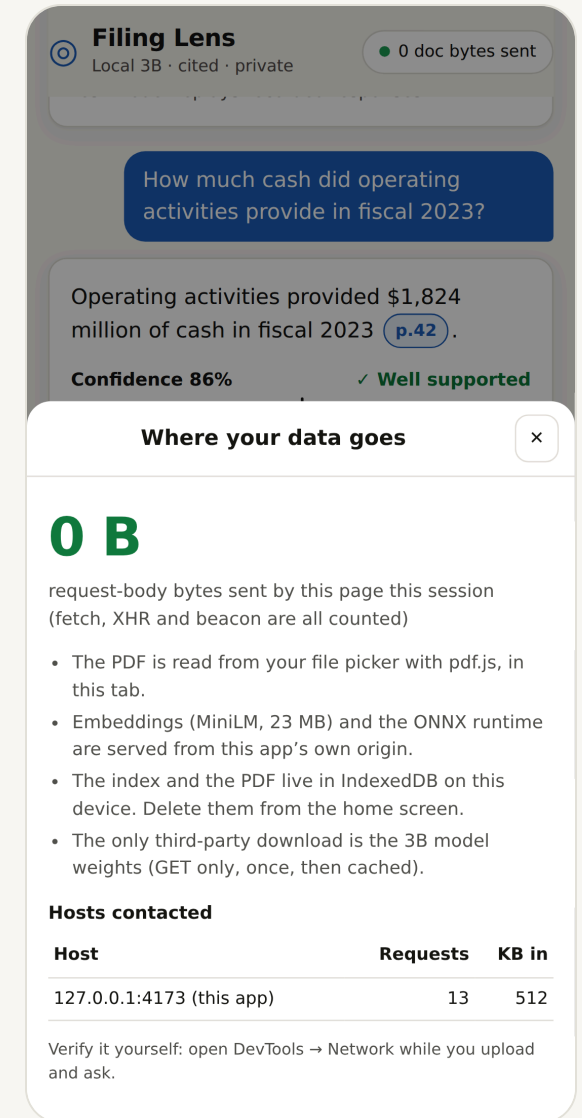
The document never leaves the device. The build's e2e test enforces this; it isn't just claimed.

Requests during upload → index → ask → abstain **20**

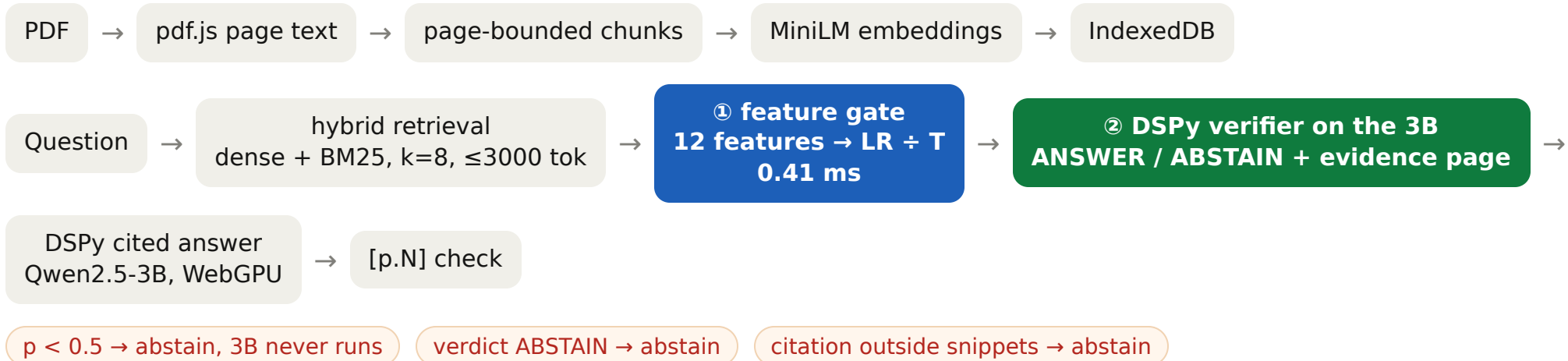
to any host other than the app's own origin **0**

carrying a request body (POST/PUT/beacon) **0**

- pdf.js, MiniLM embeddings, ONNX runtime, wllama: npm packages served from the app origin, no external scripts
- Index + PDF stored in IndexedDB
- Only third-party fetch: the Qwen2.5 weights (GET, once, then cached). Airplane mode after that: **[ASK: confirm on phone]**



Architecture: a two-stage cascade



① is cheap and catches wrong-company questions (it scores retrieval and word coverage, not content). ② reads the actual sentences: it is a DSPy program, shipped uncompiled for now (0 demos; the Kaggle compile with BootstrapFewShot, 2 demos, is pending), exported to `app/prompts/verifier.json` and rebuilt in the browser byte-for-byte. A unit test checks this against DSPy's own rendering. In the cascade, 53% of test questions reach ②.

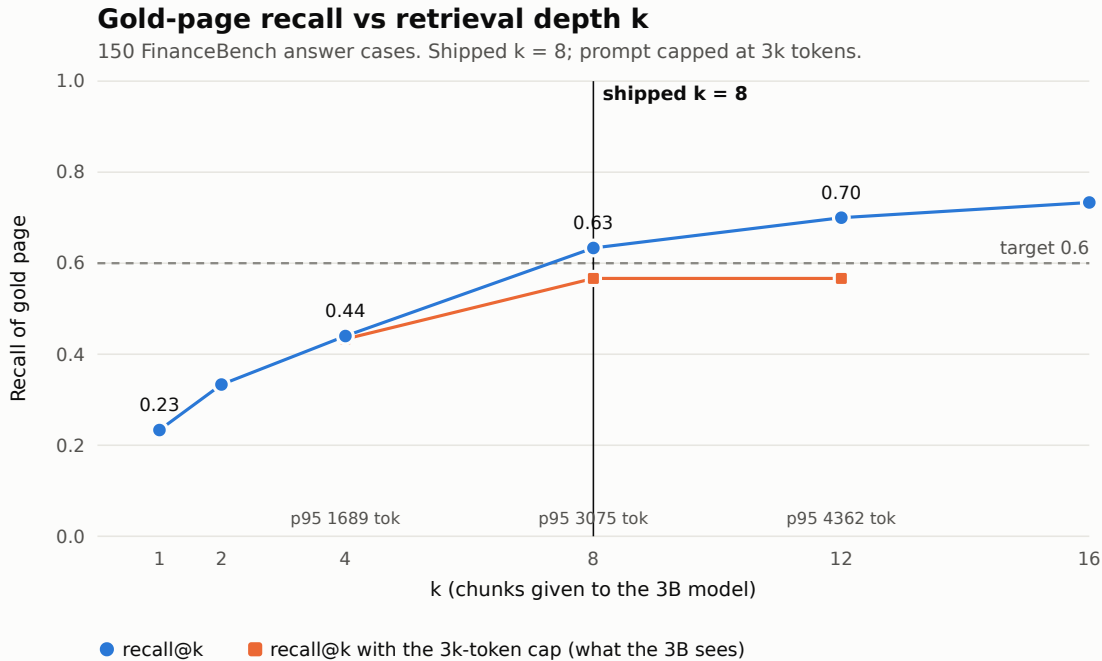
Bench: gate vs verifier vs cascade

FinanceBench open-source (patronus-ai/financebench), 150 questions, 84 filings. Gold answer = question on its own filing; **Easy** abstain = another company's 10-K; **Hard** abstain = own filing with the gold pages removed. Test split (company-grouped), 165 instances.

System	Easy acc.	Hard acc.	ECE E / H	Cite page	p50	3B calls
Feature gate only	0.864	0.518	0.191 / 0.157	0.279*	0.41 ms	0%
Verifier only (DSPy-compiled 3B)	pending (Kaggle)					100%
Cascade: gate → verifier	pending (Kaggle)					53%

* gate proposes no page: top-ranked retrieved page. Gate p50 measured in Node on Node v22.22.0, 4 vCPU container, no GPU; verifier p50 will come from the Kaggle T4 GGUF Q4 build.

Gate, all 450 out-of-fold: Easy acc 0.867, abstain P/R 0.813 / 0.953; Hard acc 0.517. Two "does the top passage answer this?" features: Hard 0.513 → 0.517.



Limits (measured)

- **The feature gate can't see content: Hard 0.518** (chance is 0.5). It scores retrieval and word overlap, so "the gold page" and "a nearby page about the same metric" look alike. The verifier stage exists for that case; its numbers are pending the Kaggle run.
- **Retrieval is the bottleneck:** gold-page recall@4 0.440, @8 0.633, @12 0.700. With the 3k-token cap the 3B sees the gold page 57% of the time; the sentence snippets keep it 53%.
- **A failure I shipped and fixed:** Best Buy's 10-K never mentions Walmart, yet "Walmart capex?" scored 0.69 and passed the first gate. Adding a feature for whether the companies a question names appear in the filing at all fixed it (gate accuracy 0.82 → 0.91 on the earlier own-vs-wrong-company bench).
- Answers are auto-graded (number within 1%, else token-F1), an approximation of FinanceBench's human grading. Scanned PDFs (no text layer) are rejected.

Roadmap

- Run the Kaggle compile; ship the compiled verifier and report verifier / cascade numbers on the GGUF Q4 build
- Retrieval: table-aware chunking for financial statements; a reranker only if one ships as an npm package (none found) and adds <150 ms
- Raise the context budget if the phone allows (uncapped k=8 prompts reach 3586 tokens)
- Indian filings: BSE/NSE annual reports, Hindi questions

The ask

[ASK: amount / what you want]

Repo: github.com/rajyyug1132/Filing-Lens-Private-Cited-Answers

Cited answers, or an honest "I can't tell". Your filing stays yours.